SEAT NO. ____________

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR (Computer Science)

SPRING SEMESTER EXAMINATIONS 2025

Time : 3 Hours **Batch 2024** **Dated : 14-JUN-25**

Max Marks : 60

Logic Design & Switching Theory - CS-251

Question 1:

a. How to cascade two 3×8 decoders to form a 4×16 Decoder? Show the enable logic and draw the block diagram to represent the connections. **[CLO-2, 6 Marks]**

b. Describe the function of 4-to-1 Multiplexer using basic logic gates. Explain how a K- map can be used to simplify the design of the combinational circuit that will be implemented using this MUX. **[CLO-1, 6 Marks]**

Question 2:

a. Design a 4 - input priority encoder when the inputs are D0, D1, D2, D3 (with D3 having the highest priority), and the inputs are D3=0, D2=1, D1=1, D0=0, what will be the encoder's output? Draw the truth table, and mention the logical expressions as well. **[CLO-2, 6 Marks]**

b. Explain the working principle of JK flip-flop, detailing its inputs (J and K) and the clock signal. Determine the state transitions (Set, Reset, Toggle and Hold). Also, compare its behavior with that of a simple SR flip flop. **[CLO-1, 6 Marks]**

Question 3:

a. Design a Decade Counter (mod-10) using JK flip-flops with reset. Draw the state diagram, truth table and the block diagram. **[CLO-2, 6 Marks]**

b. What is a shift register? Define and explain different types of Shift Registers with the help of block diagrams. **[CLO-1, 6 Marks]**

Question 4:

a. Draw the state table and generate the output sequence for 1010011 input sequence. **[CLO-1, 6 Marks]**

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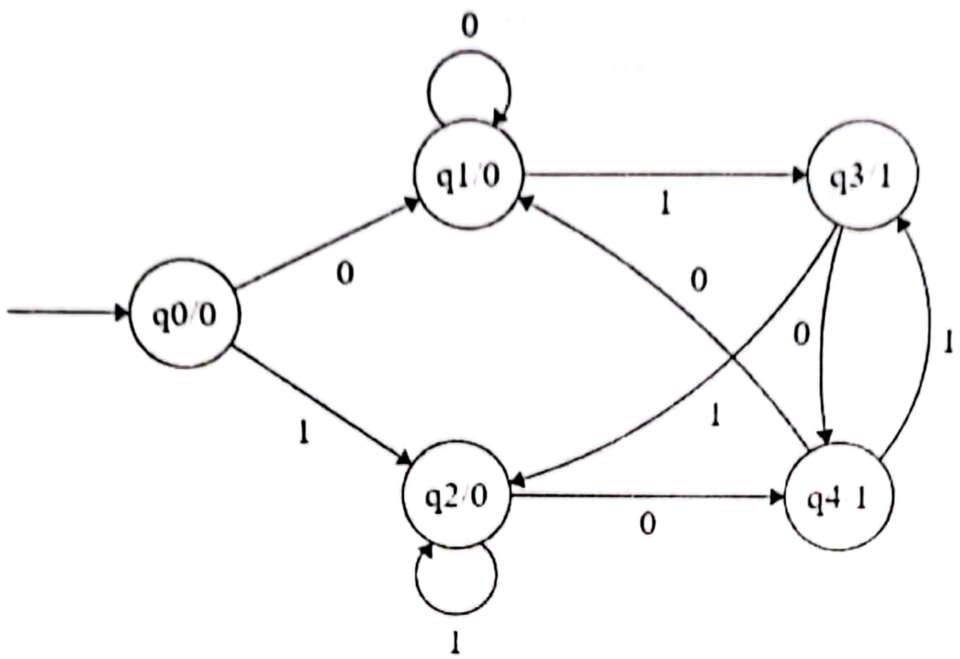

b. Design a 2-bit (Mod-4) synchronous counter using JK flip flops. **[CLO-2, 6 Marks]**

Question 5:

a. Construct SR flip flops and include all the parameters (truth table, characteristic table, characteristic equation and its excitation table). **[CLO-2, 6 Marks]**

b. Design and implement a 4-bit magnitude comparator using digital logic. Your solution should cover all three comparison cases: (A>B, A=B, A<B). Derive the Boolean expressions for each output condition (A>B, A=B, A<B). Use an example of 7485 4 bit magnitude comparator. **[CLO-1, 6 Marks]**